8 Particles X (of mass 4 units) and Y (of mass 9 units) move directly towards each other, collide and then separate. If Δv_x is the change of velocity of X and Δv_Y is the change in velocity of Y, what is the magnitude of the ratio $\frac{\Delta v_X}{\Delta v_Y}$?

A $\frac{9}{4}$ **B** $\frac{3}{2}$ **C** 1 **D** $\frac{2}{3}$